IPC PHASE 4 — SOUTH SUDAN — 2024 CURRENT PERIOD

# Food emergency drivers: flood correlation summary

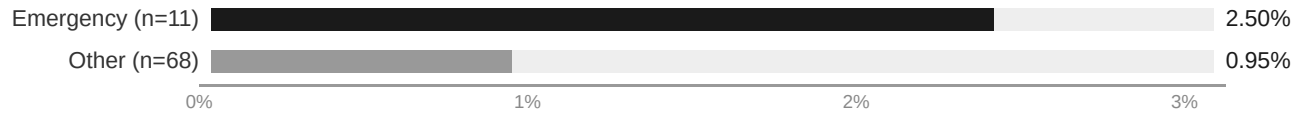
Source: IPC Phase 4 flood correlation report · Prepared for Capstone data challenge

| 11 → 43 | 3.1× | 6.0× | 34% |
|---|---|---|---|
| COUNTIES, HARVEST → LEAN SEASON | MORE FLOODED LAND, PHASE 4 COUNTIES | MORE EXTREME FLOOD EVENTS | EMERGENCY RATE AMONG RETURNEES |

## DRIVERS

- **Floods, not receding** — dominant trigger, >80% of Phase 4 counties
  - Unusual outflows from Lake Victoria / East African Great Lakes, pushed down the White Nile
  - Sudd wetlands inundated continuously, 2020–2024
  - Drowned sorghum & maize · livestock foot rot · cholera/malaria from contaminated water points · roads cut
- **Conflict + Sudan war returnees**
  - Intercommunal violence & cattle raiding block farming
  - 800,000+ refugees/returnees since 2023–24, concentrated in Renk, Malakal, Rubkona, Aweil
  - Returnees: 34% Phase 4 emergency rate
- **Currency collapse**
  - SSP sharp depreciation
  - Petrodar pipeline disrupted → export revenue down → food prices up
- **Drought — opposite mechanism, southeast only**
  - Eastern Equatoria: Kapoeta East, Kapoeta North, Lafon, Budi
  - Not flood-driven — rainfall deficit, dry spells, pasture shortage

## FLOOD EXPOSURE — PHASE 4 VS. OTHER COUNTIES




- Median flooded land, current 2024 period
- ≈3× more flooded land in emergency counties
- ≈6× more extreme ("unusual") flood events in emergency counties

## HARVEST → LEAN SEASON

- Flooding in Sep–Nov harvest destroys standing crops & food stores
- Emergency counties: 11 now → 43 projected by Apr–Jul 2025 lean season
- Statistically overwhelming difference (Mann–Whitney $p < 0.001$)

## COUNTY DETAIL

| COUNTY | STATE | POP. IN P4 | FLOODED | WATER SOURCES |
|---|---|---|---|---|
| **Rubkona** | Unity | 30% | 54.8% | Bahr el Ghazal River · Lake No · Naam River |
| **Fangak** | Jonglei | 25% | 12.3% | Bahr el Zeraf · White Nile floodplain |
| **Aweil South** | N. Bahr el Ghazal | 25% | 5.0% | Lol River · Pongo River · Jur River floodplain |

| COUNTY | STATE | POP. IN P4 | FLOODED | WATER SOURCES |
|---|---|---|---|---|
| **Panyikang** | Upper Nile | 25% | 3.3% | White Nile (west bank) · Bahr el Zeraf |
| **Pibor** | Jonglei | 35% | 2.8% | Pibor River · Lotilla River · Kangen River |
| **Uror** | Jonglei | 30% | 2.5% | Khor Geni · Khor Filus · seasonal streams |
| **Aweil East** | N. Bahr el Ghazal | 20% | 2.4% | Lol River · Kiir River (Bahr el-Arab) |
| **Malakal** | Upper Nile | 30% | 1.5% | White Nile (Bahr al-Abyad) · Sobat River |
| **Luakpiny/Nasir** | Upper Nile | 20% | 0.9% | Sobat River · Baro River · Pibor confluence |
| **Aweil North** | N. Bahr el Ghazal | 25% | 0.3% | Kiir River (Bahr el-Arab) · Lol River, upper tributaries |
| **Nyirol** | Jonglei | 25% | 0.2% | Khor Nyanding · Khor Filus, upper reaches |

Source: IPC Phase 4 flood correlation report — causal driver analysis, current-period comparison (n=11 Phase 4, n=68 non-Phase 4), county water-source listing. Lean-season figures are the Apr–Jul 2025 projection.